

```

global_data = {

    # --- Attribute Names ---
    "attribute_1": "total_cost",          # Always attribute 1
    "attribute_2": "material_continuity", # Always attribute 2; 1 = previously Cu,
    "attribute_3": "power_rating_kw",     # Rated mechanical power output (kW)
    "attribute_4": "efficiency_class",     # IEC efficiency class: IE1=1, IE2=2, IE3=
    "attribute_5": "operating_temperature_rating_c", # Max insulation temp (°C):

    # --- Directions ---
    "direction_attribute_1": "negative", # Lower cost is better
    "direction_attribute_2": "positive", # Higher material continuity with current choice is better
    "direction_attribute_3": "positive", # Higher power output is better
    "direction_attribute_4": "positive", # Higher efficiency class is better
    "direction_attribute_5": "positive", # Higher thermal tolerance is better

    # --- Generalized Min/Max (domain-wide; not dataset-specific) ---

    # Total cost: fractional-HP consumer motor (~$100) → large mult
    "attribute_1_min": 100,
    "attribute_1_max": 500000,

    # Material continuity: binary by definition
    "attribute_2_min": 0,    # Previously aluminum
    "attribute_2_max": 1,    # Previously copper

    # Power rating: small appliance motor (0.1 kW) → large industrial drive motor (10
    "attribute_3_min": 0.1,
    "attribute_3_max": 10000,

    # Efficiency class: IE1 (lowest standard) → IE5 (emerging super
    "attribute_4_min": 1,
    "attribute_4_max": 5,

    # Operating temp rating: Class Y (90°C, low-temp specialty) → Class C (
    "attribute_5_min": 90,
    "attribute_5_max": 220,

    # --- Commodity Prices ---
    "copper_price_per_ton": 9500,
    "aluminum_price_per_ton": 2500,
    "copper_price_per_kg": 9.5,
    "aluminum_price_per_kg": 2.5,
}

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regional_data = [
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# -----
# UNITED STATES
# - Efficiency-driven (DOE/EISA NEMA Premium mandate)
# - Balanced cost outlook; lifecycle cost thinking common
# - Strong Cu OEM incumbency and established supply chains
# -----
{
  "region": "United States",
  "copper_product_market_share": 0.65, # Cu dominant due to NEMA Premium
  "aluminum_product_market_share": 0.35,

  "weight_attribute_1": 0.20, # total_cost      – lifecycle thinking offsets upfront Cu
  "weight_attribute_2": 0.20, # material_continuity – entrenched Cu OEM preference
  "weight_attribute_3": 0.20, # power_rating      – balanced industrial sizing needs
  "weight_attribute_4": 0.28, # efficiency_class  – HIGHEST; DOE/EISA regulatory driver
  "weight_attribute_5": 0.12, # operating_temp   – LOWEST; controlled
},

# -----
# CHINA
# - Highly price-competitive manufacturing environment
# - GB 18613 pushing IE3 adoption; market rapidly evolving
# - Less material inertia; frequent Cu↔Al switching
# -----
{
  "region": "China",
  "copper_product_market_share": 0.55, # Slight Cu majority as IE3 mandates tighten
  "aluminum_product_market_share": 0.45,

  "weight_attribute_1": 0.28, # total_cost      – HIGHEST; intense cost competition
  "weight_attribute_2": 0.15, # material_continuity – LOW; market actively switching ma
  "weight_attribute_3": 0.22, # power_rating      – high due to scale/diversity of industry
  "weight_attribute_4": 0.22, # efficiency_class  – tied with power; GB 18613
  "weight_attribute_5": 0.13, # operating_temp   – slightly elevated for h
},

# -----
# INDIA
# - Most price-sensitive market; upfront cost dominates
# - BEE IS 12615 IE2/IE3 standards slowly gaining traction
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# - Tropical climate raises thermal stress significance
# -----
{
  "region": "India",
  "copper_product_market_share": 0.40, # Al still prevalent due to cost sensitivity
  "aluminum_product_market_share": 0.60,

  "weight_attribute_1": 0.35, # total_cost      – DOMINANT; highly price-drive
  "weight_attribute_2": 0.12, # material_continuity – LOWEST; younger industrial
  "weight_attribute_3": 0.20, # power_rating      – essential for proper industrial sizing
  "weight_attribute_4": 0.18, # efficiency_class   – moderate; BEE adoption la
  "weight_attribute_5": 0.15, # operating_temp     – ELEVATED; tropical cl
},

# -----
# EUROPE
# - Strictest global efficiency mandate (EU 2019/1781)
# - High energy costs shift focus to total cost of ownership
# - Strong material traceability and sustainability culture
# -----
{
  "region": "Europe",
  "copper_product_market_share": 0.70, # Cu dominant; IE3 mandate + energy cost ROI
  "aluminum_product_market_share": 0.30,

  "weight_attribute_1": 0.14, # total_cost      – LOWEST; TCO lens deprioritizes u
  "weight_attribute_2": 0.21, # material_continuity – HIGHEST; sustainability trace
  "weight_attribute_3": 0.20, # power_rating      – standard technical compl
  "weight_attribute_4": 0.35, # efficiency_class   – DOMINANT; EU Ecodesign IE
  "weight_attribute_5": 0.10, # operating_temp     – LOWEST; mild climate + advan
},
]

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product_data = [

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# =====
# UNITED STATES — COPPER DOMINANT
# =====
{
  "region": "United States",
  "dominant material": "cu",
  "product": "AC Squirrel-Cage Induction Motor, 15kW, Copper-Wound Stator",
  "copper_kg": 12, # Copper stator windings (magnet wire)

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"aluminum_kg":      2,    # Aluminum die-cast rotor bars
"total_mass_kg":    85,    # Includes iron core ~55kg, housing ~16kg, internals
"non_material_cost_per_unit": 420, # Labor, bearings, shaft, housing, overhead
# Computed total_cost = $420 + (12×$9.50) + (2×$2.50) = $539.00
"attribute_2_value": 1,    # material_continuity: previously copper → 1
"attribute_3_value": 15,   # power_rating_kw: 15 kW
"attribute_4_value": 3,    # efficiency_class: IE3 (NEMA Premium equivalent)
"attribute_5_value": 155,  # operating_temp_rating_c: Class F insulation (155°C)
},

# =====
# UNITED STATES — ALUMINUM DOMINANT
# =====
{
  "region":      "United States",
  "dominant material":  "al",
  "product":     "AC Squirrel-Cage Induction Motor, 15kW, Aluminum-Wound Stator",
  "copper_kg":   1,    # Minimal copper (terminals, connectors only)
  "aluminum_kg": 18,    # Aluminum stator windings + die-cast rotor bars
  "total_mass_kg": 78,  # Lighter; Al windings less dense than Cu per unit volume
  "non_material_cost_per_unit": 380, # Lower labor; Al die-casting is simpler
  # Computed total_cost = $380 + (1×$9.50) + (18×$2.50) = $434.50
  "attribute_2_value": 0,  # material_continuity: previously aluminum → 0
  "attribute_3_value": 15, # power_rating_kw: 15 kW
  "attribute_4_value": 2,  # efficiency_class: IE2 (standard efficiency)
  "attribute_5_value": 130, # operating_temp_rating_c: Class B insulation (130°C)
},

# =====
# CHINA — COPPER DOMINANT
# =====
{
  "region":      "China",
  "dominant material":  "cu",
  "product":     "AC Squirrel-Cage Induction Motor, 15kW, Copper-Wound Stator",
  "copper_kg":   12,
  "aluminum_kg": 2,
  "total_mass_kg": 85,
  "non_material_cost_per_unit": 420,
  # Computed total_cost = $539.00
  "attribute_2_value": 1,
  "attribute_3_value": 15,
  "attribute_4_value": 3,
  "attribute_5_value": 155,

```

```

},

# =====
# CHINA — ALUMINUM DOMINANT
# =====
{
  "region":          "China",
  "dominant material":  "al",
  "product":          "AC Squirrel-Cage Induction Motor, 15kW, Aluminum-Wound Stator",
  "copper_kg":        1,
  "aluminum_kg":       18,
  "total_mass_kg":     78,
  "non_material_cost_per_unit": 380,
  # Computed total_cost = $434.50
  "attribute_2_value": 0,
  "attribute_3_value": 15,
  "attribute_4_value": 2,
  "attribute_5_value": 130,
},

# =====
# INDIA — COPPER DOMINANT
# =====
{
  "region":          "India",
  "dominant material":  "cu",
  "product":          "AC Squirrel-Cage Induction Motor, 15kW, Copper-Wound Stator",
  "copper_kg":        12,
  "aluminum_kg":       2,
  "total_mass_kg":     85,
  "non_material_cost_per_unit": 420,
  # Computed total_cost = $539.00
  "attribute_2_value": 1,
  "attribute_3_value": 15,
  "attribute_4_value": 3,
  "attribute_5_value": 155,
},

# =====
# INDIA — ALUMINUM DOMINANT
# =====
{
  "region":          "India",
  "dominant material":  "al",

```

```

"product":          "AC Squirrel-Cage Induction Motor, 15kW, Aluminum-Wound Stator",
"copper_kg":        1,
"aluminum_kg":      18,
"total_mass_kg":    78,
"non_material_cost_per_unit": 380,
# Computed total_cost = $434.50
"attribute_2_value": 0,
"attribute_3_value": 15,
"attribute_4_value": 2,
"attribute_5_value": 130,
},

```

```

# =====

```

```

# EUROPE — COPPER DOMINANT

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# =====

```

```

{
  "region":          "Europe",
  "dominant material": "cu",
  "product":          "AC Squirrel-Cage Induction Motor, 15kW, Copper-Wound Stator",
  "copper_kg":        12,
  "aluminum_kg":      2,
  "total_mass_kg":    85,
  "non_material_cost_per_unit": 420,
  # Computed total_cost = $539.00
  "attribute_2_value": 1,
  "attribute_3_value": 15,
  "attribute_4_value": 3,
  "attribute_5_value": 155,
},

```

```

# =====

```

```

# EUROPE — ALUMINUM DOMINANT

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```

# =====

```

```

{
  "region":          "Europe",
  "dominant material": "al",
  "product":          "AC Squirrel-Cage Induction Motor, 15kW, Aluminum-Wound Stator",
  "copper_kg":        1,
  "aluminum_kg":      18,
  "total_mass_kg":    78,
  "non_material_cost_per_unit": 380,
  # Computed total_cost = $434.50
  "attribute_2_value": 0,
  "attribute_3_value": 15,

```

```
    "attribute_4_value":    2,  
    "attribute_5_value":    130,  
  },  
]
```